

Camille Pawlak

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Remote-sensing and spatial ecologist specializing in large-scale environmental monitoring, geospatial data integration, and ecological change detection. My research applies satellite, aerial, and field data to terrestrial, urban, and coastal ecosystems.

Education

2023- 2026	Ph.D. University of California, Los Angeles Geography Advisor: Dr. Thomas Gillespie
2021-2023	M.S. Cal Poly, San Luis Obispo Biology Advisor: Dr. Matt Ritter
2016-2020	B.S. University of California, Los Angeles Environmental Science Minors: GIS, Conservation Biology

Experience

2024 - Present - **Staff Scientist**, Urban Forest Institute
2025 - **Science Mentor**- Oceans Group - NASA SARP
2023 - **Urban Forest Analyst**, Dudek
2021-2023 - **Scientific Consultant**, Class One Arboriculture
2020-2021 - **Drone and Remote Sensing Technician**, UCLA Geography
Fall 2019 - **NASA Develop Intern**, Science, Systems and Applications Inc.
2018-2020 - **Undergraduate Research Assistant** (volunteer), Dr. David Jacobs, UCLA Ecology and Evolutionary Biology
2019-2020 - **Undergraduate Research Assistant**, Dr. Kyle Cavanaugh, UCLA Geography
2019 - **Herbarium Technician** (volunteer), UCLA
Summer 2018 - **Restoration Intern**, TreePeople

Peer-Reviewed Publications

- [13] **Pawlak, C. C.**, Yost, J., Ventura, J., Guizan, G., Arnold, S., Okin, G., Cavanaugh, K., Fricker, G. A., Ritter, M., & Gillespie, T. W. (2026). *Mapping California's Urban Forest at Scale: An Error-Adjusted Canopy Time Series for Monitoring Change*. Ecology.
<https://doi.org/10.64898/2026.05.04.722774> [Pre-print, Under Review @ PLOS ONE]
- [12] Zhang, Y., Wu, K., Wang, H., Pfannerstill, E. Y., Nagalingam, S., Coggon, M. M., Stockwell, C., Ruiz, V. M., Ho, C., Yu, Y., Novelly, J., **Pawlak, C.**, Kim, S., Warneke, C., Goldstein, A. H., & Guenther, A. B. (2026). Reconciling observed and modeled estimates of urban terpene emissions. *Nature Communications*. <https://doi.org/10.1038/s41467-026-75075-9>
- [11] Kenny, R., Johns, J., **Pawlak, C.**, Fricker, A., Yost, J., & Ritter, M. (2026). Urban trees and structure loss in the 2025 Eaton and Palisades fires. *Urban Forestry & Urban Greening*, 121, 129470.
<https://doi.org/10.1016/j.ufug.2026.129470>
- [10] Love, N. L. R., Althaus, K., Esperon-Rodriguez, M., Ossola, A., Johns, J., **Pawlak, C.**, Igić, B., Ritter, M., & Yost, J. (2025). Genetic diversity in urban forests and their potential role in conserving the world's threatened tree species. *Annals of Botany*, mcaf238.

<https://doi.org/10.1093/aob/mcaf238>

- [9] Vannucci, P. F., Kianmehr, A., **Pawlak, C. C.**, Faiola, C. L., Wilson, J. P., & Berelson, W. M. (2025). The role of urban forest composition in modulating summertime air quality in Los Angeles. *Science of The Total Environment*, 1003, 180715. <https://doi.org/10.1016/j.scitotenv.2025.180715>
- [8] Thorne, J. H., Godfrey, J. M., Boynton, R. M., Shapiro, K. D., Stern, M. A., **Pawlak, C.**, Ritter, M., & Choe, H. (2025). A spatial triage of at-risk conifer forests to support seed collection efforts and sustainable forestry. *Journal of Environmental Management*, 373, 123654. <https://doi.org/10.1016/j.jenvman.2024.123654>
- [7] Ventura, J., **Pawlak, C.**, Honsberger, M., Gonsalves, C., Rice, J., Love, N. L. R., Han, S., Nguyen, V., Sugano, K., Doremus, J., Fricker, G. A., Yost, J., & Ritter, M. (2024). Individual tree detection in large-scale urban environments using high-resolution multispectral imagery. *International Journal of Applied Earth Observation and Geoinformation*, 130, 103848. <https://doi.org/10.1016/j.jag.2024.103848>
- [6] Rendon, P., Love, N., **Pawlak, C.**, Yost, J., Ritter, M., & Doremus, J. (2024). Street tree diversity and urban heat. *Urban Forestry & Urban Greening*, 91, 128180. <https://doi.org/10.1016/j.ufug.2023.128180>
- [5] **Pawlak, C. C.***, Love, N. L. R.*, Yost, J. M., Fricker, G. A., Doremus, J. M., & Ritter, M. K. (2023). California's native trees and their use in the urban forest. *Urban Forestry & Urban Greening*, 89, 128125. <https://doi.org/10.1016/j.ufug.2023.128125>
- [4] Cavanaugh, K. C., Cavanaugh, K. C., **Pawlak, C. C.**, Bell, T. W., & Saccomanno, V. R. (2023). CubeSats show persistence of bull kelp refugia amidst a regional collapse in California. *Remote Sensing of Environment*, 290, 113521. <https://doi.org/10.1016/j.rse.2023.113521>
- [3] Saccomanno, V. R., Bell, T., **Pawlak, C.**, Stanley, C. K., Cavanaugh, K. C., Hohman, R., Klausmeyer, K. R., Cavanaugh, K., Nickels, A., Hewerdine, W., Garza, C., Fleener, G., & Gleason, M. (2023). Using unoccupied aerial vehicles to map and monitor changes in emergent kelp canopy after an ecological regime shift. *Remote Sensing in Ecology and Conservation*, 9(1), 62-75. <https://doi.org/10.1002/rse2.295>
- [2] Love, N. L. R., Nguyen, V., **Pawlak, C.**, Pineda, A., Reimer, J. L., Yost, J. M., Fricker, G. A., Ventura, J. D., Doremus, J. M., Crow, T., & Ritter, M. K. (2022). Diversity and structure in California's urban forest: What over six million data points tell us about one of the world's largest urban forests. *Urban Forestry & Urban Greening*, 74, 127679. <https://doi.org/10.1016/j.ufug.2022.127679>
- [1] Houskeeper, H. F., Rosenthal, I. S., Cavanaugh, K. C., **Pawlak, C.**, Trouille, L., Byrnes, J. E. K., Bell, T. W., & Cavanaugh, K. C. (2022). Automated satellite remote sensing of giant kelp at the Falkland Islands (Islas Malvinas). *PLOS ONE*, 17(1), e0257933. <https://doi.org/10.1371/journal.pone.0257933>

In preparation

Love, N.L.R., E.L. Grady, **C. Pawlak**, J. Yost, J. Ventura, G.A. Fricker, J. Doremus, and M. Ritter. In prep. Patterns in the distribution of public vs. private urban trees: Implications for tree equity in one of the world's largest urban forests. Target journal: *Urban Forestry & Urban Greening*.

Johns J., Love, N.L.R., **C. Pawlak**, J. Yost, and M. Ritter. In Prep. Using climatic niches to plan climate resilient urban forests. Target journal: *Urban Forestry & Urban Greening*.

C. Pawlak, R. Kenny, T. Ruiz, J. Schedler, K. Veerasingam, A. Fricker, J. Yost, and M. Ritter. In Prep. Integrating Arborist Records and Remote Sensing to Improve Urban Tree Allometric Equations and Benefit Estimates. Target journal: *Scientific Data*.

Scholarships

\$14,500	Graduate Dean's Scholar Award, University of California, Los Angeles (2023-2024)
\$12,000	David and Frieda Wertman Scholarship, Cal Poly, San Luis Obispo (2021, 2022)
\$300	Western Chapter of International Society of Arboriculture Annual Conference Student

Research Grants

\$8,000	The Britton Fund – Research Awards Program (2022) • Trends in Native Tree Planting in California
\$5,000	Cal Poly, San Luis Obispo Data Strategic Research Initiative – Grant Incubation Competition (2022) • Expansions and Utilization of the California Urban Forest Inventory

Dataset Publications

Baiza, J., **Pawlak, C.**, Baehr, A., Yost, J., Ritter, M., & Fricker, A. (2022). *Mapping Social and Environmental Justice Across California Schools (Version V4) [dataset]*. Harvard Dataverse. <https://doi.org/10.7910/DVN/7NNBJD>

Becerra, M., Rivera, O., **Pawlak, C.**, Crocker, A., & Pinto, N. (2022). *Base de datos de cobertura de cultivos de cacao en la Amazonia Peruana (Version V3) [dataset]*. Harvard Dataverse. <https://doi.org/10.7910/DVN/XMQNC2>

Reports and Professional Publications

CAL FIRE Urban and Community Forestry & Urban Forest Ecosystems Institute. (2026). *Greening the Golden State: Urban Forest Canopy Plan for California* [Draft strategic plan]. Named collaborator and contributing author.

Pawlak, C., Love, N., Yost, J., Ritter, M. (2023). Native to Where? California's Native Trees and Their Use in the Urban Environment. *Western Arborist*.

Komen, J., Falco, C., **Pawlak, C.**, & Hodel, D. (2023). Gallery Characteristics of the Invasive Shot Hole Borer and Extent of Accompanying Fusarium Dieback Disease Spread in Relation to the CODIT Model and Principles in London Plane Trees. *Western Arborist*.

Hodel, D. , Falco, C., **Pawlak, C.**, & Komen, J. (2022). Testing treatment effectiveness for Invasive Shot Hole Borers. *Western Arborist*.

Map Publications

Camille Pawlak. 2025. Maps created for California's Trees: A Field Guide to the Native Species (Authors: Matt Ritter and Michael Edward Kauffmann). Backcountry Press.

Camille Pawlak. 2023. Maps created for Vascular Plants of San Luis Obispo County, California (Author: David J. Keil). Pacific Street Publishing.

Presentations

2025 February. Tracking California's Urban Forest: Big Data and Remote Sensing for Smarter Management. Invited Talk at Climate Solutions Now – Online.

2024 December. Harnessing Neural Networks for Broad-Scale Urban Canopy Cover Mapping: A Remote Sensing Approach in California. Presentation at the American Geophysical Union annual conference – Washington D.C.

2024 May. Open Canopy: mapping California's urban canopy. Presentation at the Western Chapter of the International Society of Arboriculture annual conference – San Diego, CA.

2023 August. Data-Driven Urban Forest: How We Can Use Data and Tools to Manage Urban Forests. Invited talk at the Professional Tree Care Association of San Diego annual meeting – San Diego, CA.

2023 May. Trends in California's Native Trees in the Urban Environment. Presentation at Cal Poly College of Science & Mathematics 2023 Conference – San Luis Obispo, CA.

2023 May. Data-Driven Urban Forest: How We Can Use Data and Tools to Manage Urban Forests. Invited talk at the Annual ReLeaf Network Retreat – Sacramento, CA.

2023 May. New Tools for the Urban Forest Ecosystem Institute. Invited talk at the Western Chapter of the International Society of Arboriculture annual conference – Palisades, CA.

2022 November. Trends in California's Native Trees in the Urban Environment. Invited talk at the annual Britton Fund Putting Research to Work Conference – Colma, CA.

2022 October. Data-Driven Urban Forest: Using Data to Make Better Species Selections. Invited talk at the California Urban Forest Council: Lead with Trees Conference – Rancho Mirage, CA.

Poster Sessions

2023 August. OpenCanopy: Leveraging aerial imagery and deep learning to delineate California's urban tree canopy. SIGKDD Conference on Knowledge, Discovery, and Data Mining – Long Beach, CA.

2022 November. Trends in California's Native Trees in the Urban Environment. Poster at San Luis Obispo GIS Users Group annual GIS day – San Luis Obispo, CA

2022 November. May the forest be with you: utilizing GEDI for crop height estimations in the Amazon, Poster at San Luis Obispo GIS Users Group annual GIS day – San Luis Obispo, CA

2022 June. Toward Automatic Urban Forest Inventories with Remote Sensing. Data Strategic Research Initiative Poster Session, California Polytechnic State University– San Luis Obispo, CA

Teaching Assistantships

Spring 2025 UCLA; GEOG 7 Introduction to GIS

Winter 2025 UCLA; GEOG 5 People and Earth's Ecosystems

Fall 2024 UCLA; GEOG 109 Invasive Species

Spring 2023 Cal Poly, SLO; GEOG 440 Advanced GIS

Spring 2023 Cal Poly, SLO; BOT 433 Field Botany

Fall 2022 Cal Poly, SLO; BOT 121 Introductory Botany

Spring 2022 Cal Poly, SLO; BIO 163 Introductory Ecology and Evolution

Mentoring

2025	Logn Jewell, Grace Woerner, Melanie Lin, Alex Lacayo, John Lund, and Molly Mackellar NASA SARP Summer Research Program
2024-2025	Katharine McDaniel – Undergraduate research, Cal Poly SLO Mapping urban tree cover in California
2024	Miles Caldwell – Undergraduate summer research, Pasadena City College Tracking tree health and characteristics in Pasadena, CA
2024	April Engelmeier – Undergraduate summer research, UC Davis Mapping urban tree cover in California
2023-2024	Sara Arnold, Griffen Guizan, Ryley Chase, Lexxie Crocker, Jessica Baiza - Frost and SURP Summer Research Program, Cal Poly SLO Mapping urban tree cover in California
2022-2023	Jessica Baiza and Alice Baehr – Senior Project, Geography and Anthropology, Cal Poly SLO Mapping schoolyard canopy cover
2022-2023	Erin Grady – Frost Scholar in Biology, Cal Poly SLO Assessing the suitability of urban tree species to California's current climates
2022-2023	Cameron Gonsalves and Ben Brown – Senior Project, Geography and Anthropology, Cal Poly SLO Mapping Trees of Laguna Lake, CA
2022	Keilana Sugano and Cameron Gonsalves – Frost Summer Research Program, Cal Poly SLO Tree Detection in Urban Environments
2022	Kylee Nielson – Undergraduate research, Cal Poly SLO Canopy cover in mobile home parks
2022	Olivia Ross – Undergraduate research, Cal Poly SLO Mapping Santa Cruz Island
2022	Ben Glossner, Sarah Rietkerk, Jenna Whilden, and Eshley Freed-Doerr – Computer Science capstone, Cal Poly SLO Tracking San Luis Obispo's Trees: 10tall.calpoly.edu
2022	Ben Mangelsdorf, Ryan Zhang, and Nathan Johnson – Undergraduate data science capstone, Cal

	Poly SLO
	Bird Diversity in California Urban Forest
2020-2021	Skyler Lu – Undergraduate Research, Cal Poly SLO
	Mapping California public and private trees
2020-2023	Paola Rendon – Undergraduate Senior Project in Economics, Cal Poly SLO
	Street tree biodiversity and heat island effect

Awards

2024	Esri User Conference, 3 rd Place, Communicating Science Spatially - Urban Canopy: Mapping California's Canopy Cover from 2016-2022
2024	Award of Arboriculture Research, Western Chapter of the International Society of Arboriculture (WCISA)

Skills

Programming (Python, R, Google Earth Engine); remote sensing and GIS (ArcGis Pro, ArcGis Online, QGIS, ENVI); analyzing, creating and maintaining large dataset (Google Big Query/Google Cloud Service, AWS); botany; UAV flight and data management (Map Pilot, Agisoft Metashape, Micasense Atlas, Drone Deploy, DJI Go); UAV pilot (FAA UAV License part 107); scientific and grant writing; teaching and public speaking

Media Coverage

- Slater, E. (2024) Shaping California's Future: Cal Poly to Lead the Charge for Greener Cities. College of Engineering Connection. <https://ceng.calpoly.edu/connection/2024/11/shaping-californias-future-cal-poly-to-lead-the-charge-for-greener-cities/>
- Wilson, N. (2023) Urban Tree Diversity and Global Warming. Cal Poly Bailey College of Science and Mathematics Intersections Magazine. <https://cosam.calpoly.edu/intersections-2023/urban-tree-diversity>
- Ferreira, G. (2023). Beating the Heat: Students Map California Tree Canopy to Cool Cities as Climate Warms. Cal Poly News. <https://www.calpoly.edu/news/beating-heat-students-map-california-tree-canopy-cool-cities-climate-warms>

Service

2026	Scientific Review Committee – California Emerald Ash Borer Preparation Plan
2025-2026	Technical Contributor – California Air Resources Board Natural and Working Lands Carbon Inventory
	Provided urban forest data, technical expertise, and analytical support for statewide carbon accounting.
2024-2025	Graduate Student Colloquium Chair – Department of Geography, UCLA

Peer Reviewer

Methods in Ecology and Evolution [1], Urban Forestry and Urban Greening [1], Arboriculture and Urban Forestry [1], Annals of Geography [1]